SYNTH-05: Synthetic Network Monitoring

> **Series:** SYNTH — Synthetic Monitoring | **Notebook:** 5 of 6 | **Created:** December 2025 | **Last Updated:** 10/25/2026

Network Availability, DNS, and ICMP Monitoring

This notebook covers Dynatrace Synthetic **Network Availability Monitors** (multi-protocol monitors), which test ICMP (ping), DNS, and TCP port reachability.

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Prerequisites

- ✅ Access to a Dynatrace environment with Synthetic Monitoring
- ✅ Completed SYNTH-01 through SYNTH-04
- ✅ A private synthetic location — *"Network availability monitors are supported only on private Synthetic locations"*

⚠️ **Data model — read first:** the docs describe *"three types of network availability monitors"* — ICMP, TCP and DNS. All three are the same entity in the data model (the classic name is *multi-protocol monitor*):
- **Entity (classic):** `dt.entity.multiprotocol_monitor` — deprecated in DQL, supported for as long as Dynatrace Classic is supported
- **Entity (Smartscape, preferred):** node type `NETWORK_AVAILABILITY_MONITOR` (model `dt.smartscape.network_availability_monitor`) — note the **name changed**; it is *not* `MULTIPROTOCOL_MONITOR`
- **Events:** `fetch dt.synthetic.events | filter event.type == "multiprotocol_monitor_execution"` — the literal is unverified: the validation tenant had no NAM executions and the Synthetic events reference does not list it. Run `fetch dt.synthetic.events, from:-24h | summarize count(), by:{event.type}` on a tenant with NAM monitors and adjust the filter
- **Metric:** `dt.synthetic.multi_protocol.executions` (execution count, by `dt.entity.multiprotocol_monitor`)
- **Success/failure:** `result.state` (`SUCCESS` / `FAIL`) / `result.status.code` — same as other monitor types

The rename is the trap here. Of the four synthetic classic entities, this is the one whose Smartscape node type cannot be derived from its classic name — `multiprotocol_monitor` became `NETWORK_AVAILABILITY_MONITOR`, matching the product's UI name rather than the old internal one. `smartscapeNodes "MULTIPROTOCOL_MONITOR"` raises **no error**; it returns zero rows, which is indistinguishable from an environment that has no network monitors. If a Smartscape network-monitor query comes back empty, confirm the node type spelling before concluding you have nothing to monitor. (SaaS 1.344, released 07/27/2026 with a staged tenant rollout from 07/29/2026, is what makes `dt.smartscape.*` the primary surface for the Synthetic app — verify it has reached your tenant; the node type is queryable via DQL either way. FAQ-16 covers the general migration mechanics.)

The **protocol-specific measurements** are published as metric keys: `dt.synthetic.multi_protocol.dns.resolution_time`, `dt.synthetic.multi_protocol.icmp.round_trip_time`, `dt.synthetic.multi_protocol.icmp.packets_sent` / `.icmp.packets_received` (packet loss is the difference), `dt.synthetic.multi_protocol.icmp.success_rate` and `dt.synthetic.multi_protocol.tcp.connection_time`, alongside `dt.synthetic.multi_protocol.availability` and `dt.synthetic.multi_protocol.execution_time` at monitor level. The execution-

> event field names (`result.statistics.*`) are not listed there — run the discovery query in the next cell before building event-based latency queries.

1. Network Monitoring Overview

Synthetic Network Availability Monitors

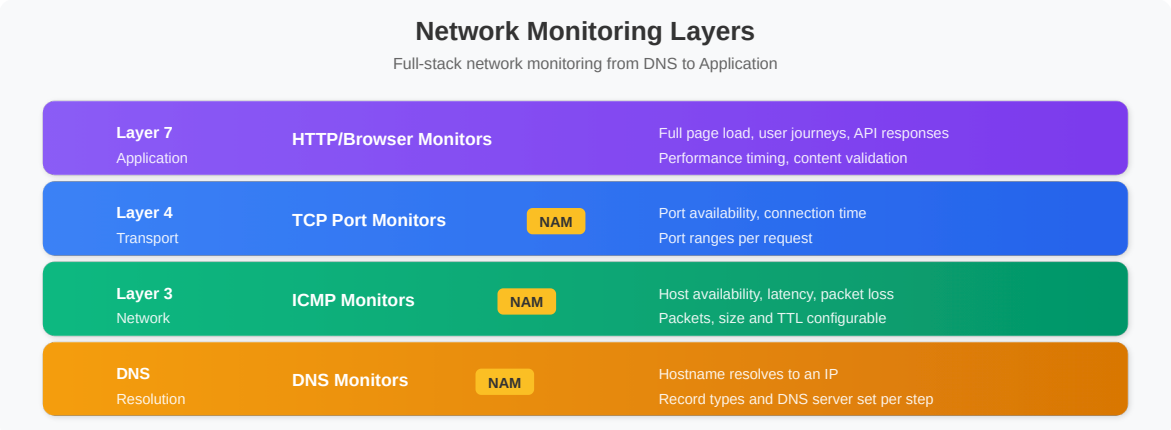
Network availability monitors come in three types:

Protocol	Purpose	Use Case
ICMP	Host reachability	Server availability
DNS	Name resolution	DNS infrastructure health
TCP	Port connectivity	Service port availability

A NAM monitor can have several steps, and *"Unlike for HTTP and browser monitors, NAM monitors can contain multiple requests within a single step. All requests assigned to a particular step are executed in parallel."* — so one ICMP monitor can ping a whole host group. Limits: up to 1,000 network activities per monitor and 5,000 NAM monitors per environment.

Why Network Monitoring?

Network monitors complement application-level monitoring by testing at different layers of the stack:



Benefits

- **Root Cause Isolation**: Distinguish network vs application issues
- **Infrastructure Validation**: Verify network paths are operational
- **DNS Health**: Monitor critical DNS infrastructure
- **Low Overhead**: Minimal resource consumption
- **High Frequency**: every 1, 2, 5, 10, 15 or 30 minutes, every hour, or on demand

> **Sources:** Network availability monitoring (DT docs) — *"There are three types of network availability monitors."* and *"The maximum number of network activities executed per network availability monitor is 1,000."*, Create a NAM monitor (DT docs) — *"All requests assigned to a particular step are executed in parallel."*

```
// DISCOVERY -- list your network availability (multi-protocol) monitors,
// then inspect the fields a recent execution emits so you know the exact
// protocol-specific result.statistics.* names available in YOUR tenant.
//
// PREFERRED -- Smartscape. The node type is NETWORK_AVAILABILITY_MONITOR, NOT
// MULTIPROTOCOL_MONITOR: the classic entity was renamed, not transliterated,
and the
// wrong spelling returns zero rows instead of an error.
smartscapeNodes "NETWORK_AVAILABILITY_MONITOR", from: now() - 30d  // the
default 2 h window can return none
| fields name, id, id_classic, network_monitor.type, enabled, frequency
| sort name asc
| limit 50

// FALLBACK (classic surface -- deprecated in DQL, supported for as long as
// Dynatrace Classic is supported):
// fetch dt.entity.multiprotocol_monitor
// | fields id, entity.name
// | sort entity.name asc
// | limit 50

// To inspect execution fields, run separately:
// fetch dt.synthetic.events, from: now() - 24h
// | filter event.type == "multiprotocol_monitor_execution"
// | limit 5
```

2. ICMP (Ping) Monitors

What ICMP Monitors Test

"ICMP—Sends pings with a configurable number of packets or size to validate if there's a network connection to the host or device. It also checks the quality of that connection."

Configuration Options

Setting	Description	Range / default
Target	IP address, hostname, or a target filter (for example a host group)	—
Number of packets	Echo requests (`ping -c` / `-n`)	1–10, default 1
Data length	Packet size (`ping -s` / `-l`)	0–65500, default 32
Time to live, **Timeout to reply**, **Do not fragment**	ICMP-only execution attributes	—
Request timeout	Per request	up to 2 minutes
Success-rate constraint	Share of requests in a step that must succeed	default ≥ 80%

Creating an ICMP Monitor

Dynatrace menu → Synthetic → Create synthetic monitor → Create network availability monitor → ICMP

Metrics Captured




```
// Network monitor execution results (last 24h) — multi-protocol events
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* — Synthetic events model, 09/21/2026.
// Covers ICMP/DNS/TCP checks; result.state reflects overall execution success
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| fields timestamp,
         monitor = monitor.name,
         location = getNodeName(dt.smartscape.synthetic_location),
         state = result.state,
         status = result.status.message,
         duration_ms = result.statistics.duration / 1ms
| sort timestamp desc
| limit 100
```

```
// Latency / duration statistics by network monitor (successful executions)
// duration is the execution time; for protocol-specific latency (e.g. ICMP
RTT,
// packet loss) add the field name found via the discovery query above.
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| filter result.state == "SUCCESS"
| summarize {
    avg_ms = avg(result.statistics.duration / 1ms),
    min_ms = min(result.statistics.duration / 1ms),
    max_ms = max(result.statistics.duration / 1ms),
    p95_ms = percentile(result.statistics.duration / 1ms, 95),
    executions = count()
  }, by: {monitor.name}
| sort avg_ms desc
| limit 20
```

3. DNS Monitors

What DNS Monitors Test

"DNS—Validates if a hostname can be resolved to an IP address."

Configuration Options

Setting	Description	Example
Hostname	Domain to resolve	`api.example.com`
DNS server	Resolver to query, optional port; system default if empty	`1.1.1.1`, `dns.google:53`
Record types	Comma-separated; each type × each host is one request	`A,AAAA`

DNS Record Types

Type	Purpose	Example
`A`	IPv4 address	192.168.1.1

Type	Purpose	Example
`AAAA`	IPv6 address	2001:db8::1
`CNAME`	Canonical name	www → app.example.com
`MX`	Mail exchanger	mail.example.com
`TXT`	Text records	SPF, DKIM
`NS`	Name servers	ns1.example.com

```
// Network monitor execution volume by monitor (metric path)
// Scope to your DNS-checking monitors by filtering
dt.entity.multiprotocol_monitor
timeseries executions = sum(dt.synthetic.multi_protocol.executions),
    from: now() - 24h, interval: 1h, by: {dt.entity.multiprotocol_monitor}
```

```
// Network monitor availability by monitor and location (events path)
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| summarize {
    total = count(),
    successful = countIf(result.state == "SUCCESS")
  }, by: {monitor.name, dt.smartscape.synthetic_location}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
2)
| fieldsAdd location = getNodeName(dt.smartscape.synthetic_location)
| sort availability_pct asc
| limit 30
```

4. TCP Port Monitors

What TCP Monitors Test

"TCP—Establishes a TCP connection to a particular port. It validates if a port is open and if it accepts TCP connections." It does not perform a TLS handshake; use an HTTP monitor for certificate checks.

Common Ports to Monitor

Port	Service	Use Case
22	SSH	Server management access
80	HTTP	Web server (plain)
443	HTTPS	Web server (secure)
3306	MySQL	Database connectivity
5432	PostgreSQL	Database connectivity
6379	Redis	Cache connectivity
9200	Elasticsearch	Search connectivity
27017	MongoDB	Database connectivity

Configuration

Setting	Description	Example
Host	Target server	`db.example.com`
Port ranges	Single ports or ranges; each port × each host is one request	`5432`, `8000-8010`

```
// Network monitor availability summary by monitor (events path)
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| summarize {
    total = count(),
    successful = countIf(result.state == "SUCCESS"),
    failed = countIf(result.state == "FAIL")
  }, by: {monitor.name}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
2)
| sort availability_pct asc
| limit 20
```

```
// Network monitor execution-volume trend (metric path, last 7 days)
timeseries executions = sum(dt.synthetic.multi_protocol.executions),
    from: now() - 7d, interval: 1h
```

5. Multi-Protocol Monitors

Combining Network Checks

Layer the three types on the same target — one monitor per type, each with its own problems and alerting:

Monitor	Type	Target	Check
1	DNS	`db.example.com`	Resolves to `10.0.1.50`
2	ICMP	`10.0.1.50`	Host reachable
3	TCP	`10.0.1.50:5432`	PostgreSQL port open

Monitoring Strategy by Layer

Layer	Monitor Type	Purpose
DNS	DNS Monitor	Name resolution works
Network	ICMP Monitor	Host reachable
Transport	TCP Monitor	Service port open
Application	HTTP Monitor	Service responding

6. Use Cases and Patterns

Infrastructure Monitoring

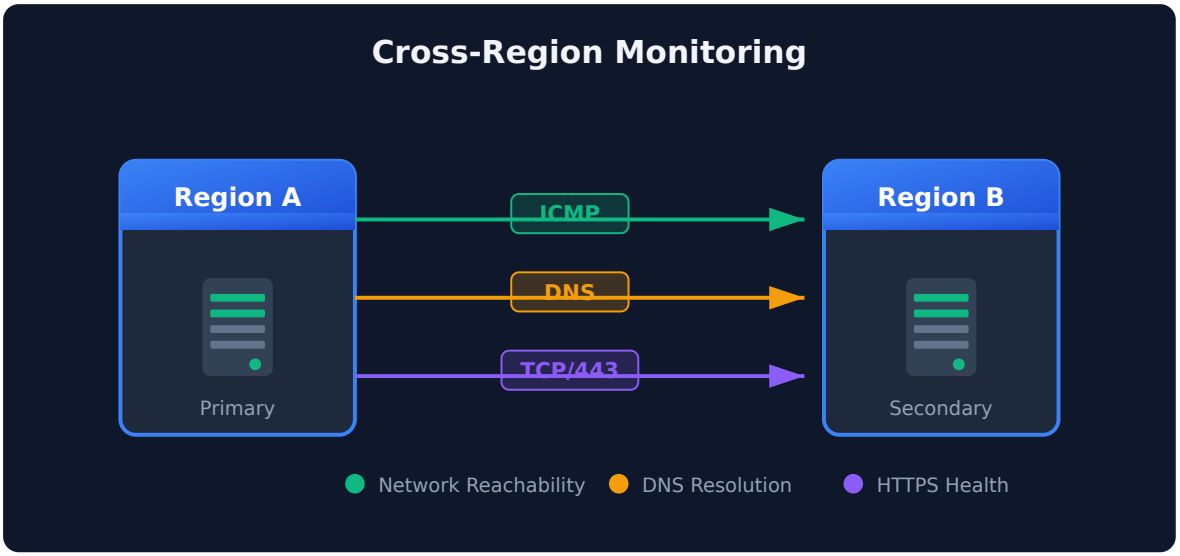
Component	Monitor Type	Target
Load Balancer	TCP/443	VIP address

Component	Monitor Type	Target
Database Cluster	TCP/5432	Each node
Cache Layer	TCP/6379	Redis instances
Message Queue	TCP/5672	RabbitMQ nodes

DNS Infrastructure

Scenario	Configuration
Primary DNS	Query internal DNS server
Secondary DNS	Query backup DNS server
External DNS	Query public resolvers
Record validation	Verify expected IP

Multi-Region Connectivity



```
// All synthetic monitor types summary (HTTP / browser / network)
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| filter not(coalesce(execution.retry_on_error, false) and result.state ==
"FAIL")  // one result per scheduled run
| summarize {
    total_executions = count(),
    successful = countIf(result.state == "SUCCESS"),
    failed = countIf(result.state == "FAIL")
  }, by: {event.type}
| fieldsAdd availability_pct = round((successful * 100.0) / total_executions,
decimals: 2)
| sort total_executions desc
```

7. Analyzing Network Results

```
// Network monitor availability over time (last 7 days)
fetch dt.synthetic.events, from: now() - 7d
| filter event.type == "multiprotocol_monitor_execution"
| makeTimeseries {
    success_count = countIf(result.state == "SUCCESS"),
    total_count = count()
  }, interval: 1h, by: {monitor.name}
| fieldsAdd availability_pct = success_count[] * 100.0 / total_count[]
```

```
// Duration trend for network monitors (last 24h)
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| filter result.state == "SUCCESS"
| makeTimeseries {
    avg_ms = avg(result.statistics.duration / 1ms),
    p95_ms = percentile(result.statistics.duration / 1ms, 95)
  }, interval: 15m
```

```
// Failed network checks with details
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| filter result.state == "FAIL"
| fields timestamp,
         monitor = monitor.name,
         location = getNodeName(dt.smartscape.synthetic_location),
         status = result.status.message,
         detail = result.status.details
| sort timestamp desc
| limit 50
```

```
// Network health score by monitor
fetch dt.synthetic.events, from: now() - 24h
| filter event.type == "multiprotocol_monitor_execution"
| summarize {
    total = count(),
    successful = countIf(result.state == "SUCCESS"),
    avg_ms = avg(result.statistics.duration / 1ms)
  }, by: {monitor.name}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
2)
| fieldsAdd avg_ms = round(avg_ms, decimals: 1)
| fieldsAdd health_status = if(availability_pct &gt;= 99.9, "HEALTHY",
                            else: if(availability_pct &gt;= 99.0, "DEGRADED",
                            else: "CRITICAL"))
| sort availability_pct asc
| limit 30
```

Summary

In this notebook, you learned:

✅ **Network availability monitors** - ICMP, DNS and TCP, one type per monitor
✅ **The data model** - `event.type == "multiprotocol_monitor_execution"` on `dt.synthetic.events`; metric `dt.synthetic.multi_protocol.executions`; entity `dt.entity.multiprotocol_monitor` (classic) or Smartscape node

`NETWORK_AVAILABILITY_MONITOR` (preferred)
✅ **The rename trap** - the Smartscape node is `NETWORK_AVAILABILITY_MONITOR`, not `MULTIPROTOCOL_MONITOR`; a wrong node type returns zero rows, not an error
✅ **Discovery first** - confirm protocol-specific `result.statistics.*` fields in your own tenant
✅ **Multi-protocol patterns** - Comprehensive infrastructure monitoring
✅ **Analysis queries** - Availability, duration, executions, failures, health score

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Next Steps

Continue to **SYNTH-06: Analytics & Alerting** to learn about dashboards, SLOs, and alerting strategies.

References

- Network availability monitoring (DT docs)
- NAM monitor metrics in Synthetic on Grail (DT docs)
- Synthetic on Grail (DT docs)
- Synthetic app (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.